

Face2Feel: Real-Time Emotion Detection and AI-Based Emotional Support System

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Abstract—Face2Feel is an AI-powered mobile counseling application that integrates real-time facial emotion detection with interactive conversational support to provide adaptive, empathetic, and private mental health assistance. The system leverages ViT-based models for emotion recognition and MediaPipe for accurate face detection, processing live camera input through a FastAPI backend. Detected emotions are fed to a conversational AI using OpenAI’s API, which generates context-aware counseling responses displayed through a Flutter-based mobile interface. By combining visual computing with AI-driven dialogue, Face2Feel enables seamless emotion-aware interactions, enhances accessibility for users hesitant to seek traditional counseling, and supports early detection of emotional distress, offering immediate guidance in diverse real-world conditions.

I. INTRODUCTION

Emotional awareness and support have become increasingly important in today’s fast-paced society, where stress, anxiety, and other mental health challenges are on the rise. Many individuals find it difficult to express their feelings openly, either due to possible social judgment, personal hesitation, or lack of immediate access to professional counseling. Mobile applications that leverage AI have the potential to bridge this gap by providing real-time feedback, personalized support, and guidance, helping users better understand and manage their emotions. Face2Feel is developed with this goal in mind: it captures user emotions from live camera input and responds through an conversational AI counselor, creating a supportive environment. The project focuses on ease of, and the seamless integration of facial emotion recognition with AI-driven counseling, ensuring that the application can provide meaningful, empathetic interactions for users across a variety of real-world scenarios.

II. RELATED WORK

Existing work in affective computing and AI-based counseling includes:

- Mobile mental health apps like Woebot and Wysa that utilize conversational AI for emotional support [2].
- Studies demonstrating the effectiveness of facial emotion recognition for understanding user states in controlled settings [1].

However, combining seamless real-time emotion detection with AI-based personalized counseling on a mobile platform remains underexplored, which *Face2Feel* addresses.

A. Mobile Application Interface

The Face2Feel mobile application is built using Flutter, offering a responsive and intuitive interface that guides users through an interactive counseling experience. The design focuses on simplicity, accessibility, and real-time engagement, ensuring that users can navigate the app without any technical expertise. The app is structured into several interconnected screens, each serving a distinct purpose:

1. Splash Screen: The application opens with a splash screen designed to create a welcoming environment. Beyond aesthetics, it provides a brief moment for initial background processes to load, ensuring a smooth transition into the main app functionality.

2. Permission Handler Page: To function as expected, Face2Feel requires access to the device camera and storage. This page prompts users to grant necessary permissions. By explicitly handling permissions upfront, the app maintains user trust, prevents runtime errors, and ensures that subsequent features operate properly.

3. Chat and Emotion Detection Screen: This is the core interface where the main functionality comes to play. The camera captures live facial expressions, and MediaPipe is used for accurate face detection. Emotions are then analyzed using a ViT-based model, and the detected emotion is displayed in a bounding box along with a confidence score. This screen also has the AI-driven chat interface, where users can communicate with the counseling bot. To enhance emotional awareness, the app dynamically adjusts the chat screen’s background color and theme based on the detected emotion, creating an immersive environment that responds in real time to the user’s emotional state.

4. Real-Time Feedback and Personalization: By combining emotion recognition with conversational AI, the app delivers context-aware feedback. Users receive immediate responses tailored to their emotional cues, and visual indicators help them understand their emotional patterns. This not only encourages honest expression but also promotes self-awareness and reflective thinking. The continuous feedback loop between emotion detection and chat interaction ensures that the counseling experience feels personal, responsive, and supportive.

5. History and Analytics Page: (Planned) This section tracks user interactions, emotional trends over time, and AI counseling sessions. It allows users to reflect on their emo-

tional patterns, providing insights into triggers and improvements in emotional well-being.

6. Settings and Personalization:(Planned) Users can configure notifications, select preferred themes, and manage privacy settings. This ensures the app is accessible and personalized according to individual preferences, while maintaining data privacy and security.

Overall, the Flutter frontend prioritizes usability, accessibility, and real-time responsiveness, creating a cohesive ecosystem where emotion detection seamlessly integrates with AI counseling. Each page is designed to guide users through self-awareness, reflection, and supportive interactions, making mental health resources more approachable and engaging.

B. Backend Processing

The backend of Face2Feel is designed to handle user inputs efficiently and provide real-time, context-aware counseling responses. It integrates a FastAPI server with the emotion recognition and AI chat modules to ensure smooth and responsive operation on mobile devices.

1. FastAPI Server: The server exposes two primary endpoints, `/predict` and `/chat`. The `/predict` endpoint receives images captured by the Flutter app, processes them through the ViT-based emotion recognition model, and returns the predicted emotion along with a confidence score. Images are first converted to RGB format and aligned using MediaPipe landmarks to ensure accurate emotion detection. The `/chat` endpoint accepts the user's text message along with the detected emotion and generates empathetic counseling responses using the OpenAI GPT-4o-mini API. If the API key is unavailable or an error occurs, predefined fallback responses ensure uninterrupted user interaction.

2. Real-Time Workflow: The backend orchestrates the interaction between the mobile frontend and AI components as follows:

- 1) The Flutter app captures a live camera image and sends it to the `/predict` endpoint.
- 2) The server aligns and preprocesses the face image, then predicts the emotion using the ViT Small model.
- 3) The predicted emotion is sent back to the app, updating the UI dynamically, including the chat screen theme to reflect the detected emotion.
- 4) When the user sends a message, it is forwarded to the `/chat` endpoint along with the current emotion.
- 5) The server constructs a context-aware prompt and queries the OpenAI API to generate an empathetic counseling response.
- 6) The response is returned to the Flutter app and displayed in the chat interface in real time.

3. Efficiency and Reliability: The backend uses asynchronous routes and lightweight computations to maintain responsiveness even on low-end devices. Robust error handling ensures fallback responses are provided in case of API errors or unexpected inputs. Detailed logging captures information about incoming requests, detected emotions, and generated responses for debugging and monitoring.

Overall, the backend enables a seamless integration of facial emotion detection and AI-driven counseling, ensuring that users receive immediate, empathetic support while interacting with the mobile application.

C. AI Counseling Module

The AI Counseling Module forms the core interactive component of Face2Feel, bridging emotion recognition with personalized conversational support. Once the user's emotion is detected by the ViT-based model and sent from the Flutter app, the backend FastAPI server uses this emotion along with the user's message to generate contextually aware counseling responses.

Workflow:

- 1) The Flutter frontend sends the detected emotion and user text message to the backend via the `/chat` API endpoint, handled by `ChatService`.
- 2) The FastAPI server constructs a system prompt for the OpenAI GPT-4o-mini model, embedding both the emotion and empathetic counseling guidelines. Each emotion has a tailored prompt to ensure responses are supportive, validating, and contextually relevant.
- 3) The OpenAI API returns a 2–4 sentence response in a soft, validating tone, often including open-ended questions to encourage user reflection.
- 4) If the API key is unavailable or an error occurs, predefined fallback responses ensure continuous counseling without interrupting the user experience.
- 5) The generated response is sent back to the Flutter app in real time, updating the chat interface and maintaining a dynamic, emotion-aware theme.

This module ensures that users receive immediate, empathetic guidance based on their current emotional state. The integration of real-time emotion detection with AI-driven conversation allows Face2Feel to simulate human-like counseling interactions while remaining fully automated and accessible on mobile devices.

D. System Architecture

Figure 3 illustrates the system architecture, highlighting the integration between the mobile front-end, backend server, and AI module. Figure 2 illustrates the flow of data

E. Model Architecture

Figure 1 illustrates the model architecture. It is the ViT Small variant which we have used, which is further fine-tuned to suit our use case.

F. Emotion Detection Module

The emotion recognition component of Face2Feel consists of two main parts: the training pipeline, which produces a compact and accurate classifier, and the real-time inference module, which processes incoming face images and predicts emotions. Both stages are built using widely adopted deep learning tools to ensure reproducibility, portability, and ease of deployment.

1) *Design Objectives*: The system was designed around three central goals:

- **Robustness to Lighting and Orientation**: Images collected in real-world conditions vary significantly, so the pipeline includes face alignment and contrast enhancement to reduce the effect of inconsistent lighting and tilted faces.
- **Handling Class Imbalance**: Emotion datasets are naturally skewed (e.g., many more “happy” or “neutral” images). To avoid the model biasing toward majority classes, the training process uses techniques that give minority expressions a more balanced influence.
- **Lightweight Deployment**: A small Vision Transformer (ViT-S) model is used with grayscale input to reduce computational cost while maintaining competitive accuracy.

2) *Training Pipeline*: The training pipeline focuses on preparing the data effectively and optimizing the model in a stable and balanced manner.

a) *Data Preparation*: Images from FER-2013 and AffectNet are first aligned using facial landmarks and stored in a unified format. During training, these images undergo controlled data augmentation (such as flips and small rotations) to help the model generalize to natural variations. Validation and test images use only standardized resizing and normalization.

b) *Class Imbalance Mitigation*: To ensure that minority emotions (such as “disgust” or “surprise”) are learned properly:

- The loss function is adjusted to assign more importance to underrepresented classes.
- The sampling strategy ensures that each training batch contains a balanced mix of emotions.

Together, these strategies greatly reduce bias and improve performance across all seven emotion categories.

c) *Model and Optimization*: A pretrained small Vision Transformer is adapted to work with single-channel (grayscale) images. This choice offers a good balance between model size and prediction accuracy. The model is trained using modern optimization methods with learning rate scheduling and early stopping to prevent overfitting.

d) *Evaluation*: After training, the final model is evaluated on a held-out test set. The metrics include precision, recall, F1-score, and confusion matrices for each emotion category, enabling a detailed understanding of strengths and failure modes.

3) *Real-Time Inference Pipeline*: A separate inference module processes incoming frames (e.g., from a webcam or mobile client) and produces emotion predictions in real time.

a) *Initialization*: The system loads the trained model and initializes the face detection and facial landmark estimation components. The same preprocessing steps used during training—such as resizing and normalization—are applied here to keep inputs consistent.

b) *Face Alignment*: To improve robustness and reduce variability caused by head tilt or uneven framing, faces are aligned before classification. This is done by estimating the

eye positions and rotating the image so that the eyes lie horizontally. A tight crop is then extracted around the facial region. This step significantly boosts accuracy, especially in non-ideal capture conditions.

c) *Image Enhancement*: Before feeding the aligned face into the model, local contrast enhancement (CLAHE) is applied. This makes the system more reliable under poor lighting, shadows, or low-contrast environments.

d) *Inference*: For each detected and aligned face, the system computes a probability distribution over the seven emotion classes. The emotion with the highest probability is selected as the final prediction. Confidence scores and optional visual overlays (like bounding boxes and predicted labels) are used during testing to inspect performance interactively.

e) *Developer diagnostics and outputs*: The script shows a combined display: webcam feed and aligned crop side-by-side, with textual debug information such as per-class probabilities and alignment failure markers (so misalignments or failed meshes are easily spotted). This interactive feedback is valuable during dataset curation, model debugging, and threshold selection for confidence-based behaviors.

4) *Why these design decisions were made*:

- **MediaPipe for landmarks**: provides robust, real-time landmark detection across devices and is lightweight compared to some alternatives. Using dense landmarks enables accurate alignment rather than naive bounding-box resizing.
- **Grayscale input (single channel)**: reduces model input size and emphasizes shape/texture cues critical for facial expression, while minimizing color variation effects.
- **Class balancing (weights + sampler)**: FER datasets tend to be heavily imbalanced (e.g., many neutral/happy samples and fewer disgust/fear examples). Combining sample-level balancing and loss re-weighting improves minority-class representation during training.
- **Moderate augmentations**: preserve expression-defining facial patterns while exposing the model to lighting and geometric variability.
- **Model choice (ViT family via timm)**: ViT-like architectures offer a good trade-off between representational power and implementation simplicity. Using `timm` enables easy switching between backbones if further experiments are required.

5) *Limitations and practical considerations*:

- **Dataset bias**: public FER datasets have inherent biases (pose, ethnicity, age distribution). Reported accuracy should be interpreted in that context; real-world performance may vary.
- **Alignment failures**: in extreme occlusion or profile poses, MediaPipe may fail to produce landmarks; the pipeline currently skips prediction for such frames — a fallback strategy (lower-confidence prediction or user prompt) can be added.

6) *Suggestions for improvement and future work*:

- **Multi-modal fusion:** combine facial expressions with speech/text sentiment for improved accuracy in ambiguous cases.
- **Domain adaptation / fine-tuning:** collect small in-domain datasets (diverse faces, lighting) and fine-tune the model using few-shot techniques to reduce domain shift.
- **Quantization / distillation:** use model compression to reduce inference cost for on-device deployment.
- **Robust fallback:** provide graceful UI prompts when alignment fails or confidence is low, for example asking the user to re-position their face.
- **Privacy-preserving inference:** explore on-device lightweight models or encrypted transmission to increase user privacy.

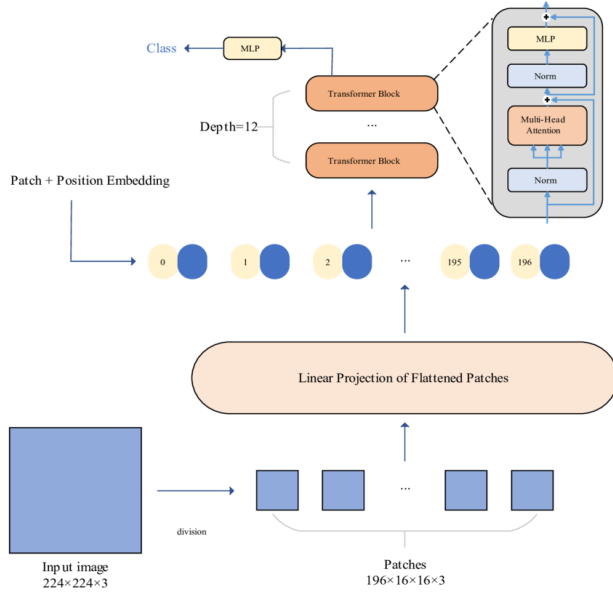


Fig. 1. ViT-Small Architecture

III. EXPERIMENTAL RESULTS

Figure 5 shows sample outputs, where the app provides emotion feedback and appropriate AI counseling.

Table I shows that the emotion recognition model achieved an overall accuracy of 68.43% across 10,192 test samples, demonstrating strong performance on expressive classes like Happy ($F1 = 0.89$) and Neutral ($F1 = 0.74$), which usually have clearer facial cues. Performance was moderate for emotions with subtle or ambiguous features, such as Anger, Fear, and Sad, each achieving $F1$ -scores between 0.52–0.62, while Disgust and Surprise also maintained reasonable recognition performance. The macro and weighted averages ($F1 = 0.65$ and 0.68, respectively) indicate that the model generalizes fairly well across imbalanced classes. Overall, the Vision Transformer demonstrates strong capability in learning global facial patterns, but subtle or visually similar emotions remain more challenging and could benefit from additional fine-tuning or data balancing.

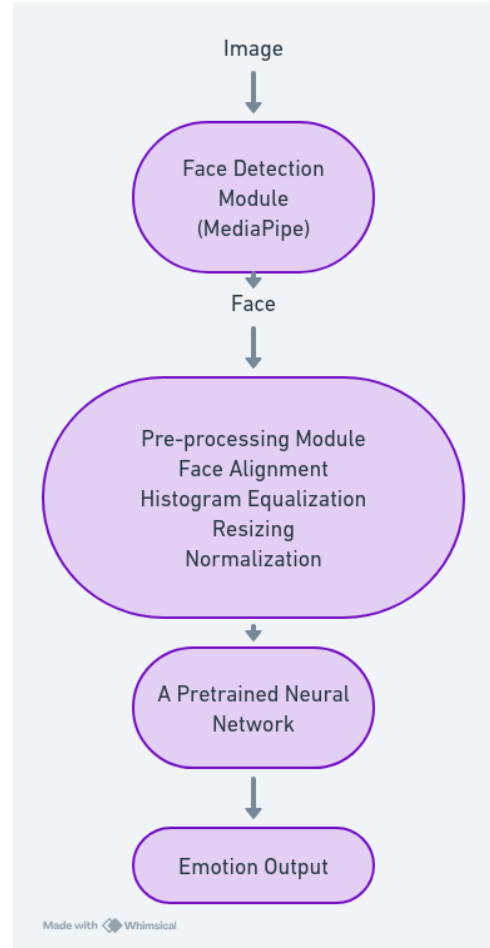


Fig. 2. Data flow for the model

Figure 4 shows the Confusion Matrix on the test dataset which contains 10,000+ images.

Class	Precision	Recall	F1-Score	Support
Anger	0.6022	0.5084	0.5513	1310
Disgust	0.4968	0.6610	0.5672	702
Fear	0.6215	0.4612	0.5295	1314
Happy	0.8988	0.8838	0.8912	2211
Neutral	0.7018	0.8001	0.7477	1841
Sad	0.6018	0.6392	0.6200	1433
Surprise	0.6476	0.6481	0.6478	1381
Accuracy		0.6843		10192
Macro Avg	0.6529	0.6574	0.6507	10192
Weighted Avg	0.6859	0.6843	0.6815	10192

TABLE I

CLASSIFICATION REPORT FOR THE EMOTION RECOGNITION MODEL.

IV. CONCLUSION AND FUTURE WORK

Face2Feel demonstrates the potential of integrating real-time emotion recognition with AI-based counseling in a mobile application. The app successfully promotes self-awareness, provides immediate emotional support, and creates an interactive environment that adapts to user emotions through changes in interface themes and conversational re-

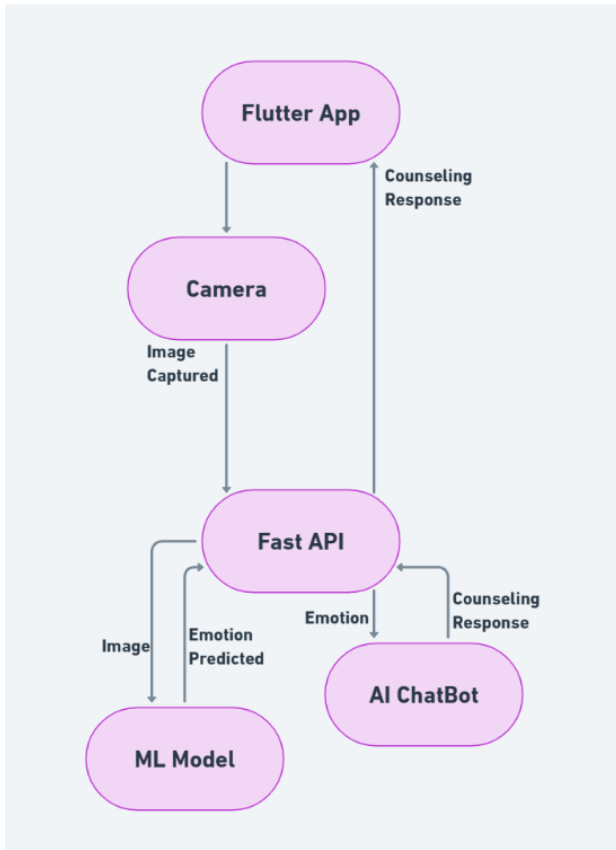


Fig. 3. Complete Software Architecture of Face2Feel.

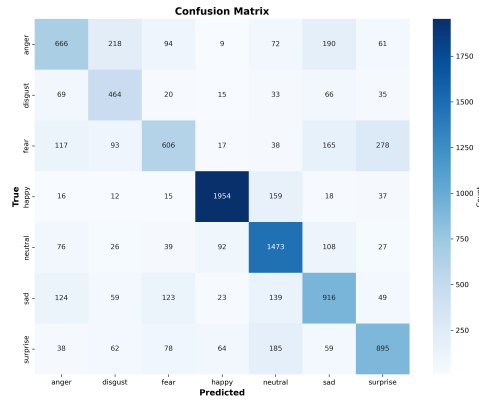


Fig. 4. Confusion Matrix of the Test set

sponses. While the current implementation provides a working prototype, there is considerable scope for improvement.

Future enhancements include:

- Integration of voice and text analysis to provide multi-modal emotional insights, enabling the app to better understand subtle cues beyond facial expressions.
- Improvements in the AI chatbot to generate more contextually nuanced and empathetic responses, enhancing the quality of counseling.
- Enhancing the accuracy and robustness of the emotion

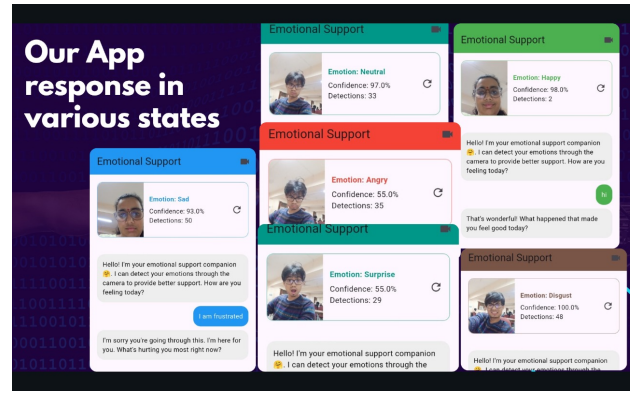


Fig. 5. Sample real-time emotion feedback and AI counselor interaction.

detection model across diverse lighting conditions, facial variations, and demographics.

- Refinement of the mobile interface with smoother animations, accessibility features, and richer visual feedback to improve user experience.
- Addition of new features, such as mood tracking over time, goal-setting for emotional well-being, and personalized suggestions for stress management or relaxation exercises.
- Conducting long-term studies to evaluate the effectiveness of the app in improving user mental health and emotional resilience.
- Incorporation of personalization mechanisms to adapt the counseling style based on user preferences, history, and behavioral patterns.

These improvements would help transform *Face2Feel* from a prototype into a more comprehensive and reliable tool for AI-driven emotional support.

REFERENCES

- [1] S. Li and W. Deng, "Deep facial expression recognition: A survey," *IEEE Transactions on Affective Computing*, 2018.
- [2] K. K. Fitzpatrick, A. Darcy, and M. Vierhile, "Delivering cognitive behavior therapy to young adults with symptoms of depression and anxiety using a fully automated conversational agent (Woebot): A randomized controlled trial," *JMIR Mental Health*, 2017.

APPENDIX

Contributor Roles Taxonomy (CRediT):

- **Shravani Naik:** Conceptualization, Methodology, Software Development, Experimentation, Writing – original draft.
- **Kshetrimayum Abo:** Model Training, Model Testing, Experimentation with Models